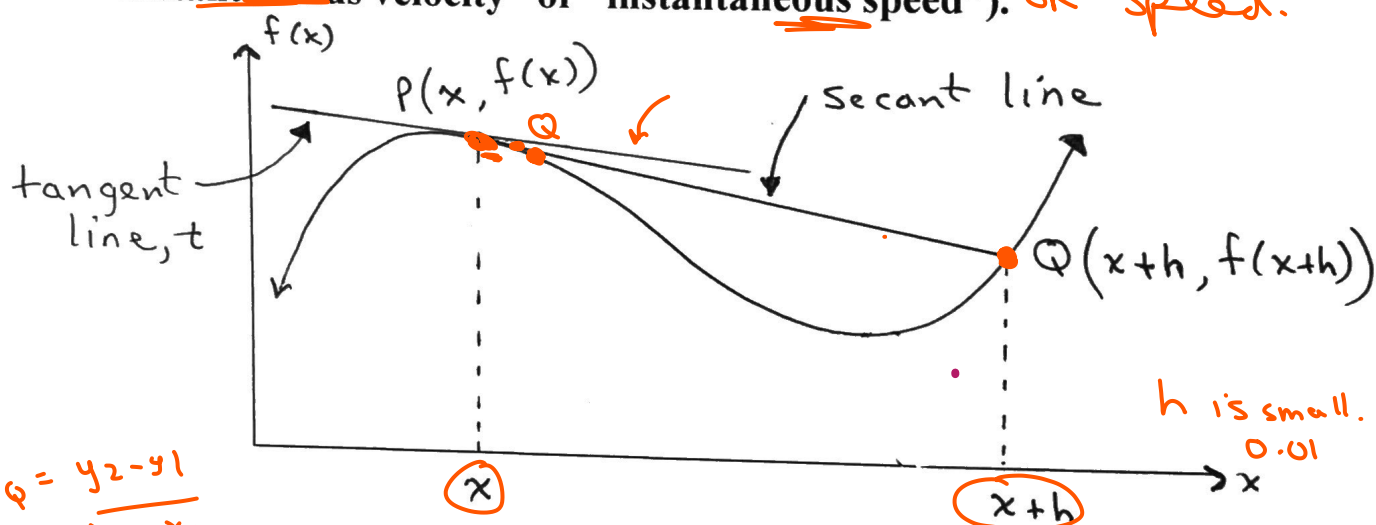


Slopes of Secants and Tangents

- The slope of the secant is also the average rate of change (AROC). (From an application point of view it is often referred to "average velocity" or "average speed").

- The slope of the tangent is also the instantaneous rate of change (IROC). (From an application point of view it is often referred to "instantaneous velocity" or "instantaneous speed"). OR "speed."



Slope of the secant line PQ is given by

$$m_{PQ} = \frac{f(x+h) - f(x)}{h}$$

Sometimes called the "Newton Quotient" or the "difference quotient".

Slope of the tangent line is given by

$$m_t = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Handwritten notes: $h=0$ and $\frac{0}{0}$ with arrows pointing to the limit process.

Example 1: Determine the slope of the tangent to the curve

$$f(x) = x^2 + 1 \text{ at the point where } x = 2.$$

$$m_t = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$m_t = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 1 - (x^2 + 1)}{h}$$

$$m_t = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 1 - x^2 - 1}{h}$$

$$m_t = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \quad \therefore m_t = 2x$$

$$m_t|_{x=2} = 4$$

$$m_t = 2x$$

Example 2: Determine the equation of the tangent to the curve

$$f(x) = \sqrt{x+3} \text{ at the point where } x = 1.$$

$$m_t = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{x+h+3} - \sqrt{x+3}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(\sqrt{x+h+3} - \sqrt{x+3})(\sqrt{x+h+3} + \sqrt{x+3})}{h(\sqrt{x+h+3} + \sqrt{x+3})}$$

$$= \lim_{h \rightarrow 0} \frac{x+h+3 - (x+3)}{h(\sqrt{x+h+3} + \sqrt{x+3})}$$

$$= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h+3} + \sqrt{x+3}}$$

$$m_t = \frac{1}{2\sqrt{x+3}}$$

$$m_t = \lim_{h \rightarrow 0} \frac{\sqrt{1+h+3} - \sqrt{1+3}}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{h+4} - \sqrt{4} - 2}{h}$$

foil

$$f(x) = \sqrt{x+3}$$

$$\therefore m_t|_{x=1} = \frac{1}{4}$$

Equation:

$$y - y_1 = m(x - x_1)$$

$$y - 2 = \frac{1}{4}(x - 1)$$

Example 3: At what point on the curve $f(x) = -2x^4$ is the tangent line perpendicular to the line $x - y + 1 = 0$?

Note: the line that is perpendicular to the tangent line at the point of tangency, P is called the "normal line".

* $x - y + 1 = 0 \Rightarrow y = x + 1 \quad \therefore \underline{m = 1} ; m_t = -1$
 (negative reciprocal)

* $m_t = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

~~m_t~~ $= \lim_{h \rightarrow 0} \frac{-2(x+h)^4 + 2x^4}{h}$

~~m_t~~ $= \lim_{h \rightarrow 0} \frac{-2[(x+h)^4 - x^4]}{h}$

~~m_t~~ $= \lim_{h \rightarrow 0} \frac{-2[(x+h)^2 - x^2][(x+h)^2 + x^2]}{h}$

~~m_t~~ $= \lim_{h \rightarrow 0} \frac{-2[\cancel{x^2} + 2x\cancel{h} + \cancel{h^2} - \cancel{x^2}][\cancel{h}][(x+h)^2 + x^2]}{h}$

~~m_t~~ $= \lim_{h \rightarrow 0} \frac{-2[2x+h][(x+h)^2 + x^2]}{1}$

~~m_t~~ $= \lim_{h \rightarrow 0} \frac{-2[2x+h][(x+h)^2 + x^2]}{1}$

$m_t = -8x^3$

But $m_t = -1 \Rightarrow -8x^3 = -1$

$x^3 = \frac{1}{8}$

$x = \frac{1}{2}$

$\therefore$ point is $(\frac{1}{2}, -\frac{1}{8})$
 $f(\frac{1}{2})$